A-09 OUTPUT Roll no : 7314

```
arinshukla@Arins-MacBook-Air ai % ./a9
=== DFS Maze Solving ===
Enter rows and columns: 4 4
Enter the maze using S, G, X, and O:
SOOX
OXOX
OXXX
OOOG

Initial maze:
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 0)

Step-by-step DFS trace:

Step 1: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 0)

Step 2: DFS moves forward to (0, 1).
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 0)

Step 3: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 1)
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Step 4: DFS moves forward to (0, 2).
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 1)

Step 5: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 2)

Step 6: DFS moves forward to (1, 2).
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 2)

Step 7: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (1, 2)

Step 8: Dead end found; DFS backtracks to this cell.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 2)

Step 9: Dead end found; DFS backtracks to this cell.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 1)

Step 10: Dead end found; DFS backtracks to this cell.
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 0)

Step 11: DFS moves forward to (1, 0).
S O O X
O X O X
O X X X
O O O G
Robot position: (0, 0)

Step 12: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (1, 0)

Step 13: DFS moves forward to (2, 0).
S O O X
O X O X
O X X X
O O O G
Robot position: (1, 0)

Step 14: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (2, 0)

Step 15: DFS moves forward to (3, 0).
S O O X
O X O X
O X X X
O O O G
Robot position: (2, 0)

Step 16: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 0)

Step 17: DFS moves forward to (3, 1).
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 0)

Step 18: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 1)

Step 19: DFS moves forward to (3, 2).
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 1)

Step 20: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 2)

Step 21: DFS moves forward to (3, 3).
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 2)

Step 22: DFS visits this cell and marks it visited.
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 3)

Step 23: Goal reached. DFS returns through the call stack.
S O O X
O X O X
O X X X
O O O G
Robot position: (3, 3)

Step 24: This cell is part of the solution path.
S O O X
O X O X
O X X X
O O * G
Robot position: (3, 2)

Step 25: This cell is part of the solution path.
S O O X
O X O X
O X X X
O * * G
Robot position: (3, 1)

Step 26: This cell is part of the solution path.
S O O X
O X O X
O X X X
* * * G
Robot position: (3, 0)

Step 27: This cell is part of the solution path.
S O O X
O X O X
* X X X
* * * G
Robot position: (2, 0)

Step 28: This cell is part of the solution path.
S O O X
* X O X
* X X X
* * * G
Robot position: (1, 0)

Step 29: This cell is part of the solution path.
S O O X
* X O X
* X X X
* * * G
Robot position: (0, 0)

Maze solved. * marks the DFS path.
S O O X
* X O X
* X X X
* * * G
Robot position: (3, 3)
arinshukla@Arins-MacBook-Air ai %